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Naive algorithm for Pattern Searching

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Given a **text** string of length **n** and a **pattern** of length **m**, find all starting indices where the pattern occurs in the text.

Note: You may assume that $n > m$.

Examples:

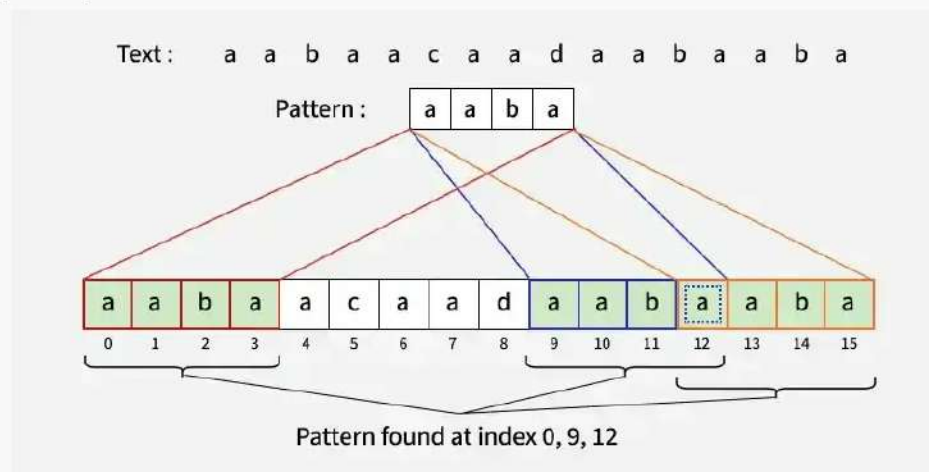
Input: text = "geeksforgeeks", pattern = "geeks"

Output: [0, 8]

Explanation: The string "geeks" occurs at index 0 and 8 in text.

Input: text = "aabaacaadaabaaba", pattern = "aaba"

Output: [0, 9, 12]



Naive Pattern Searching Algorithm

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C++ Java Python C# JavaScript

```
import java.util.*;

class Solution {

    public static List<Integer> search(String text, String pattern) {

        List<Integer> result = new ArrayList<>();

        int n = text.length();
        int m = pattern.length();

        // Check every possible starting position
        for (int i = 0; i <= n - m; i++) {

            int j;

            // Compare pattern with text
            for (j = 0; j < m; j++) {
                if (text.charAt(i + j) != pattern.charAt(j)) {
                    break;
                }
            }

            // If all characters matched
            if (j == m) {
                result.add(i);
            }
        }

        return result;
    }

    public static void main(String[] args) {

        String text = "geeksforgeeks";
        String pattern = "geeks";

        List<Integer> result = search(text, pattern);

        System.out.println(result);
    }
}
```

Output

0 9 12

Best Case Scenario

The best case occurs when the first character of the pattern does not match any character in the text.

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